

## (a) Turn-Level Agentic Tool-Calling Decision Post-Training

### Scenario in Current Turn



I need it in **Bangkok** with a good rating.



Tool Set:

**Scrambled**

get\_api\_tokens

get\_weather

.....

**Relevant**

get\_service\_providers:

```
{
  "province_id": {
    "type": "integer",
    "description": "ID of the province, with specific allowed values. Just an example, 1 for Bangkok, 2 for Chiang Mai, etc.",
    "avg_rating": "float",
    "is_cleaning_home": "boolean",
    ...
  }
}
```



Ground Truth Decision - Request for Information  
Could you please provide me with the specific province ID for Bangkok?



Agent Tool-Calling Decision



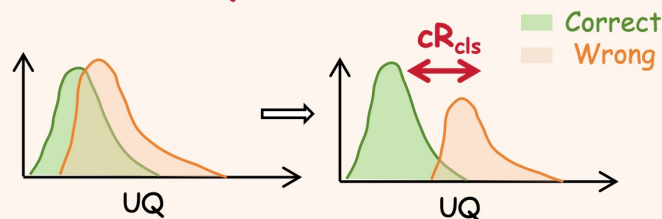
### UQ-Aligned Decision Reward $R_{UQ}$

$$R_{UQ} = R_{fmt} + R_{ans} + cR_{cls}$$

(<think> ... </think>) <Decision> <answer> ... </answer>

Consistency between Ground Truth and Agent Response

**Repulsive Force to Align  
UQ and Correctness**



### Performance



Since the example says province\_id 1 is Bangkok, the assistant should use 1...  
{... "province\_id": 1, ...}



Provide me with the specific province ID for Bangkok please?

**Task:** Find a service provider that provides cleaning services.

Multi-turn Trajectories

Turn 1

Turn 2

... Turn n

Recognized as Key Turn



### Json Annotation

```
{
  "turn_idx": 2,
  "gt_action": "Ask",
  "rationale": "Missing province ID...",
  "target_answer": "Could you please provide me with the specific province ID for Bangkok?"
}
```

Trajectory Augmentation



Cold Start SFT



RL Training



Tool Simulation



Checklist Labeling →  $R_{CM2}$



Filtered Json Annotation →  $R_{UQ}$



LLM as Judge

Multi-turn Trajectories

Post-Training Pipeline

## (b) Trajectory-Level Unified Post-Training